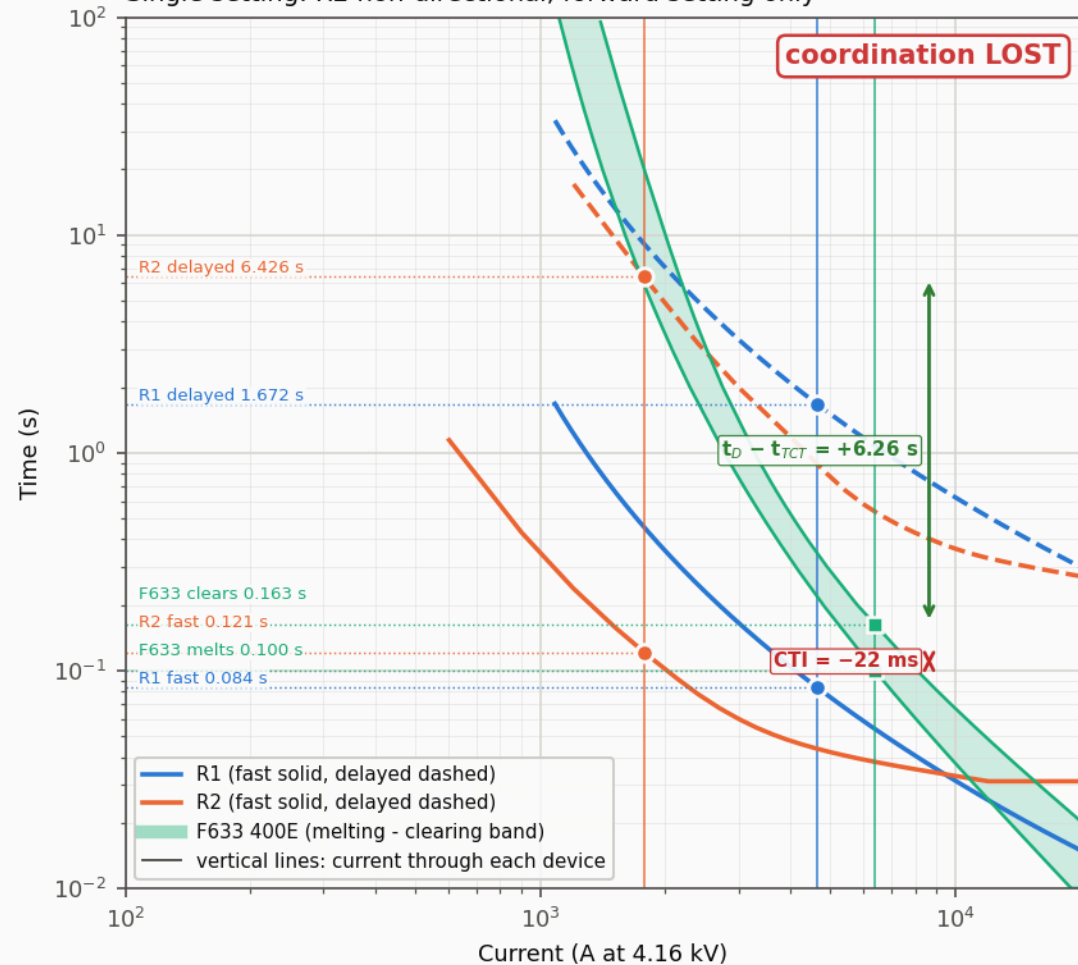
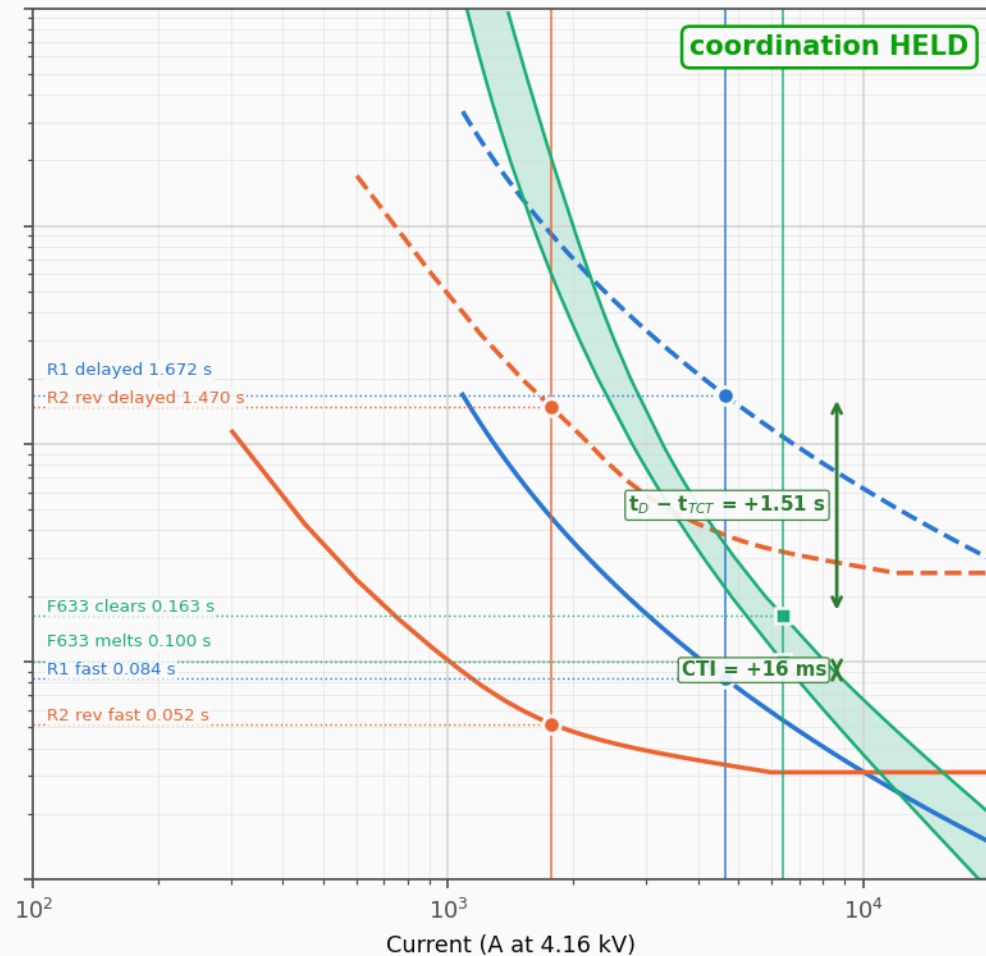


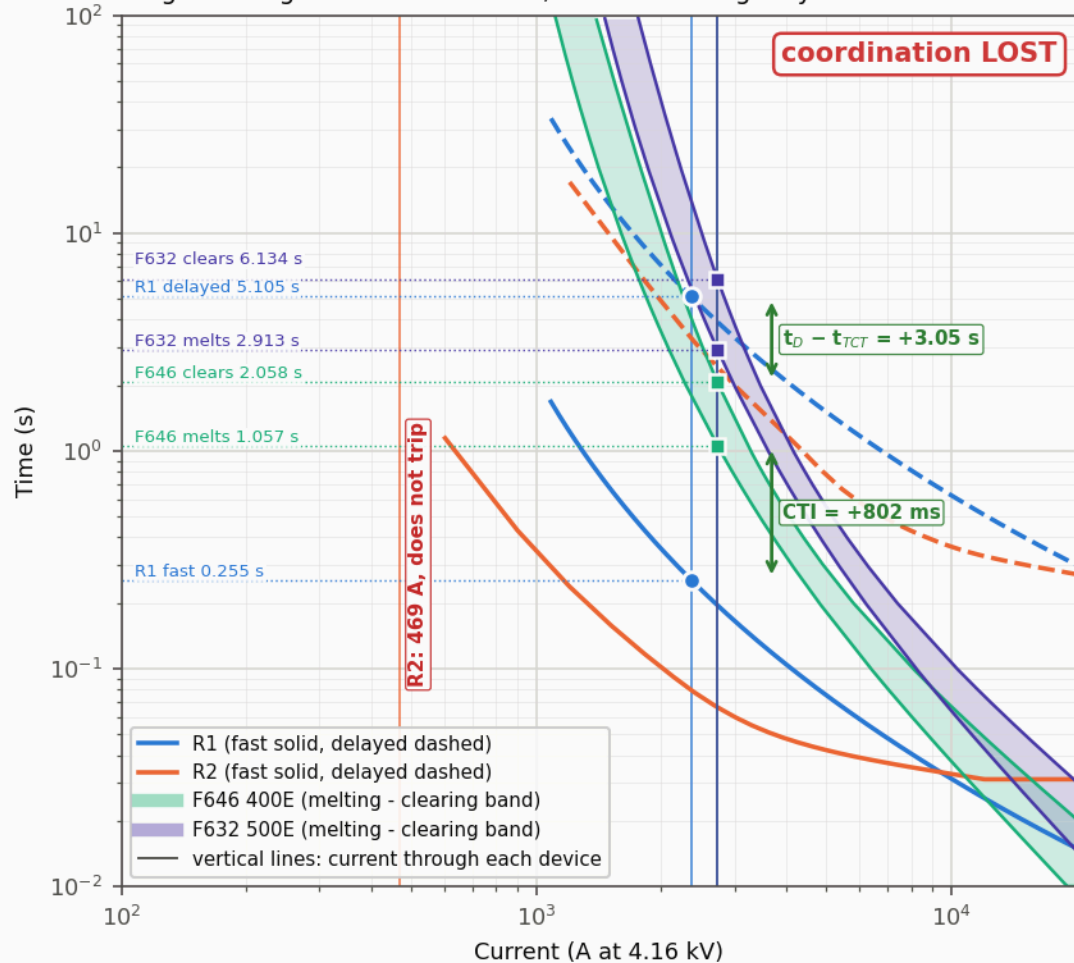
Single setting: R2 non-directional, forward setting only



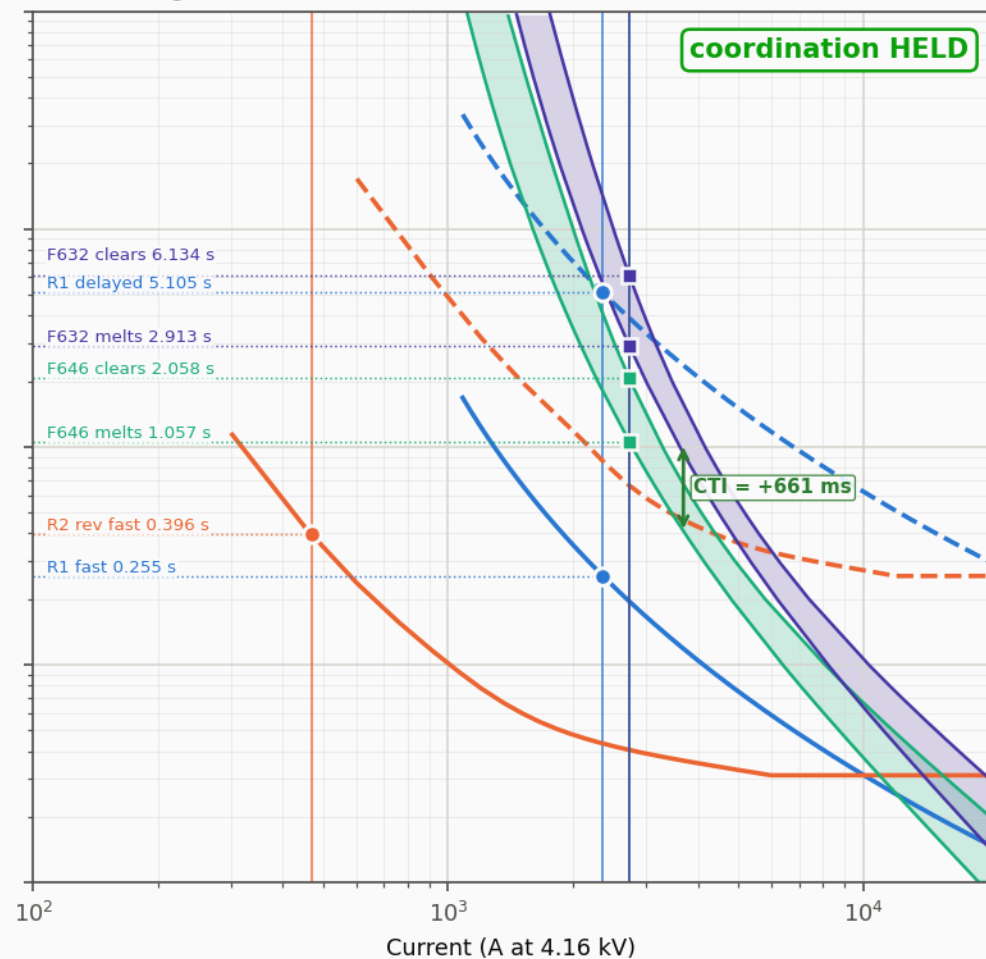
Dual setting: R2 as DSDR



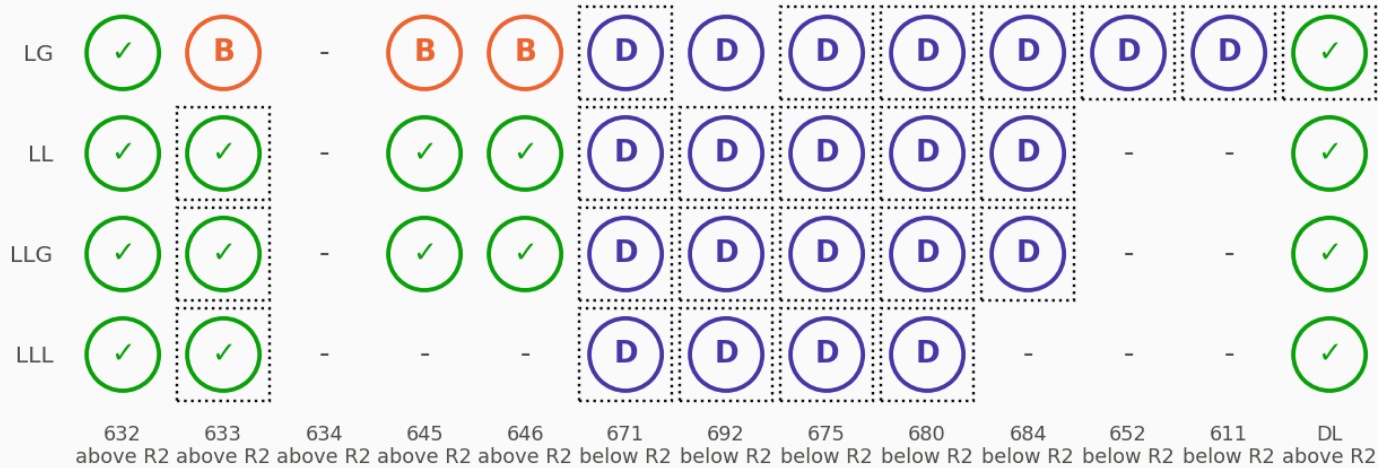
Single setting: R2 non-directional, forward setting only



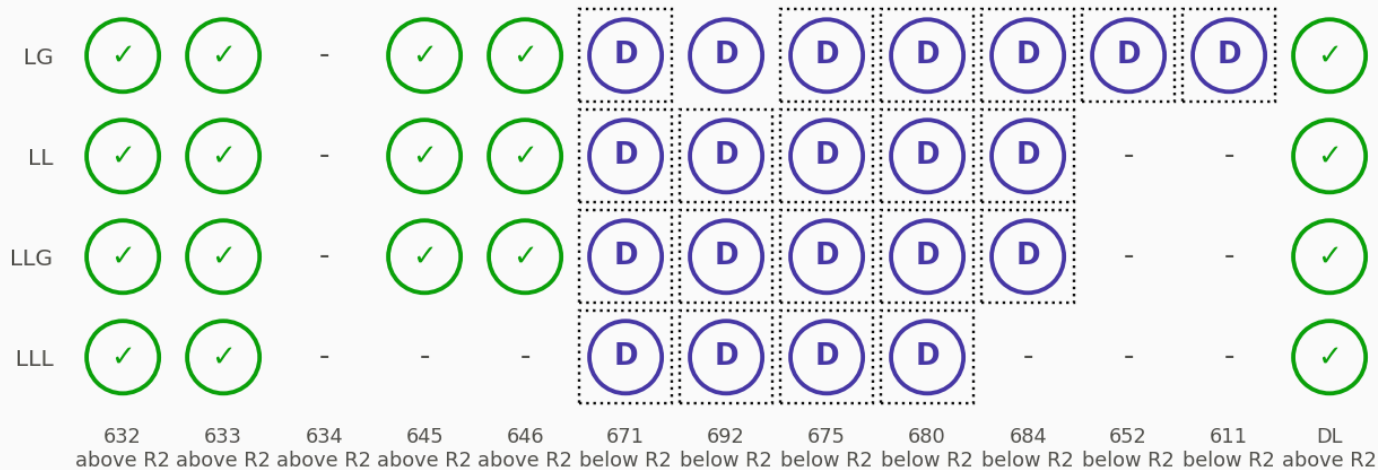
Dual setting: R2 as DSDR



Single setting on R2 - time sequence: above R2 15 of 18 held; below R2 the DG keeps feeding 21 of 21



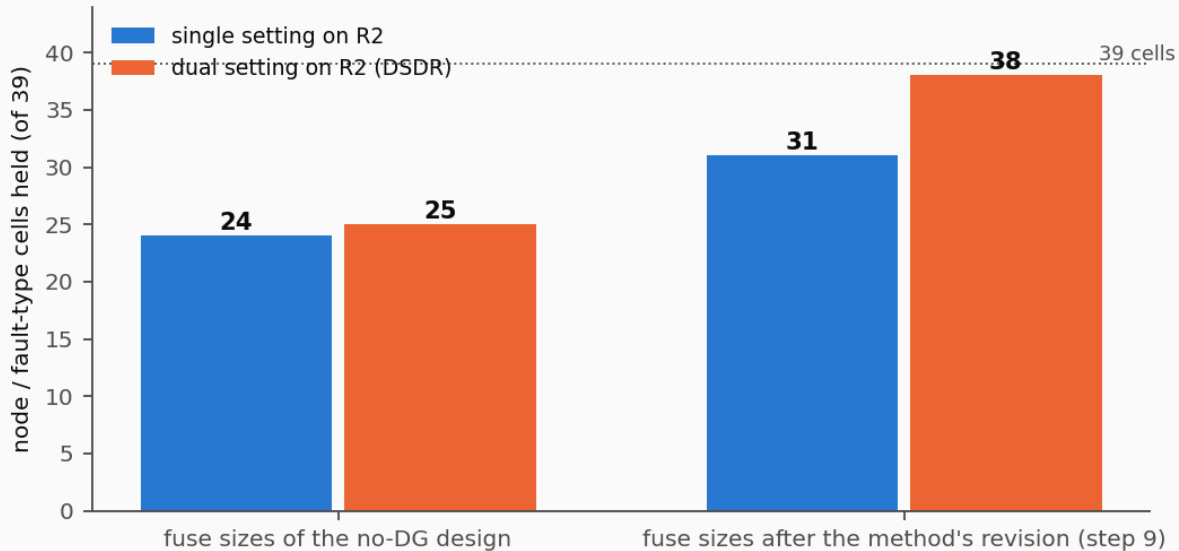
Dual setting on R2 (DSDR) - time sequence: above R2 18 of 18 held; below R2 the DG keeps feeding 21 of 21



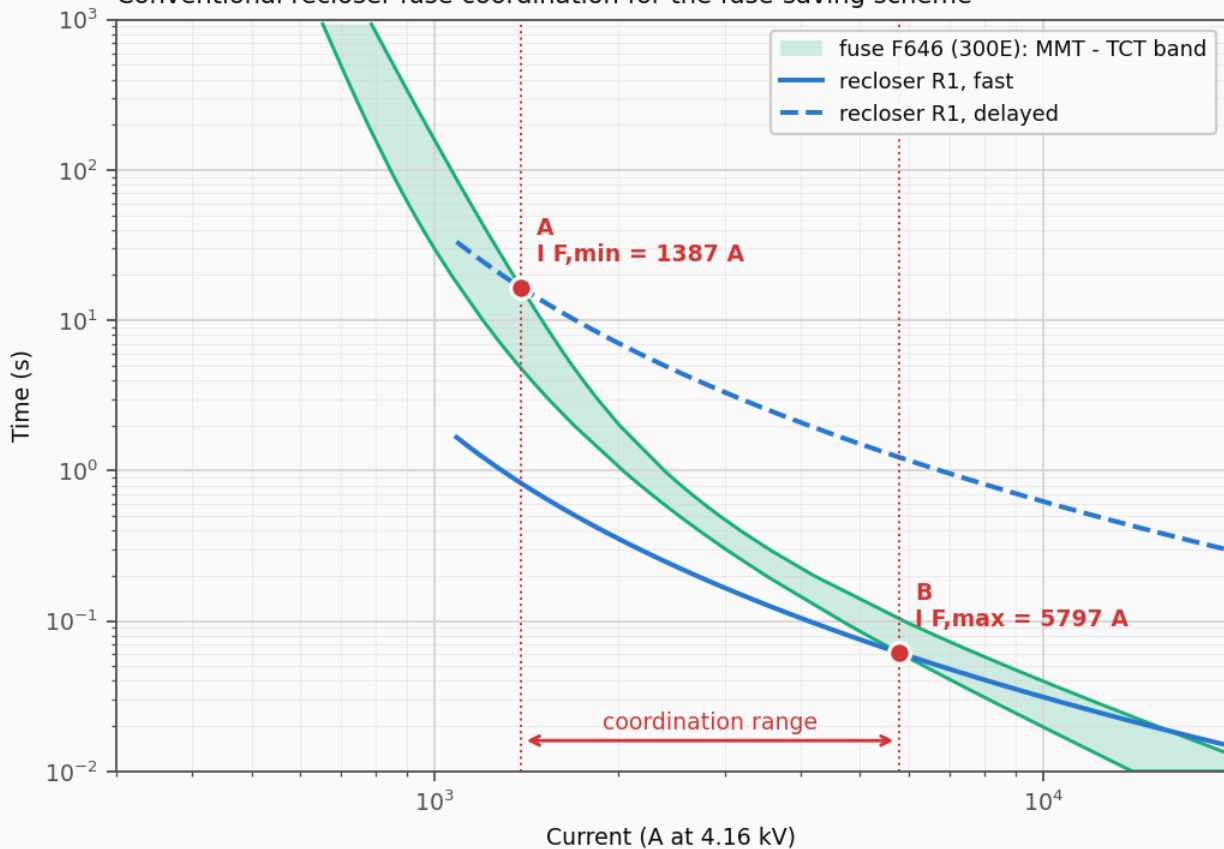
✓ held: fault cut off, fuse below 100 % heat
 O fast sequence fine, a delayed-trip / series condition fails
 D below R2: the DG keeps feeding the fault

B R2 does not trip on the DG's reverse current
 X fuse melts before the reclosers finish
 ... differs from the strict rule

Fuse-saving coordination with the DG connected



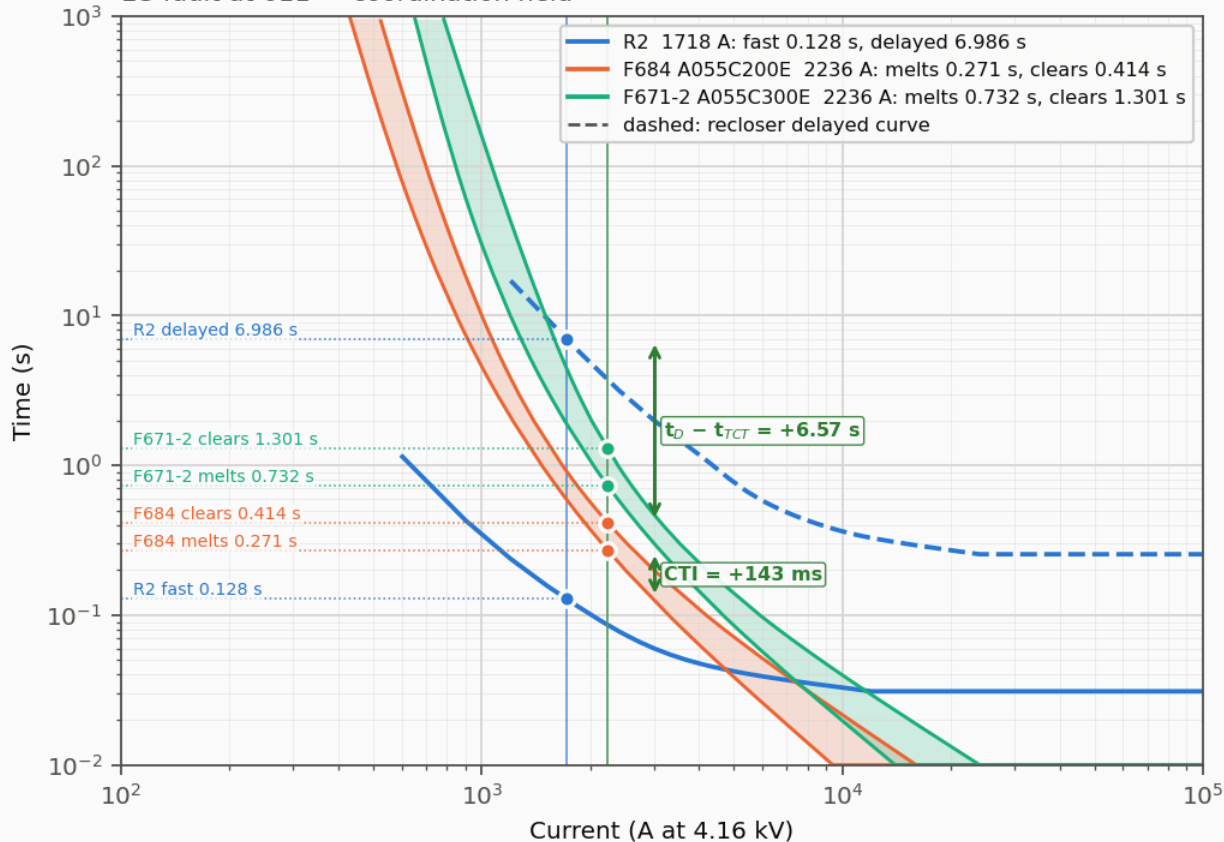
Conventional recloser-fuse coordination for the fuse-saving scheme



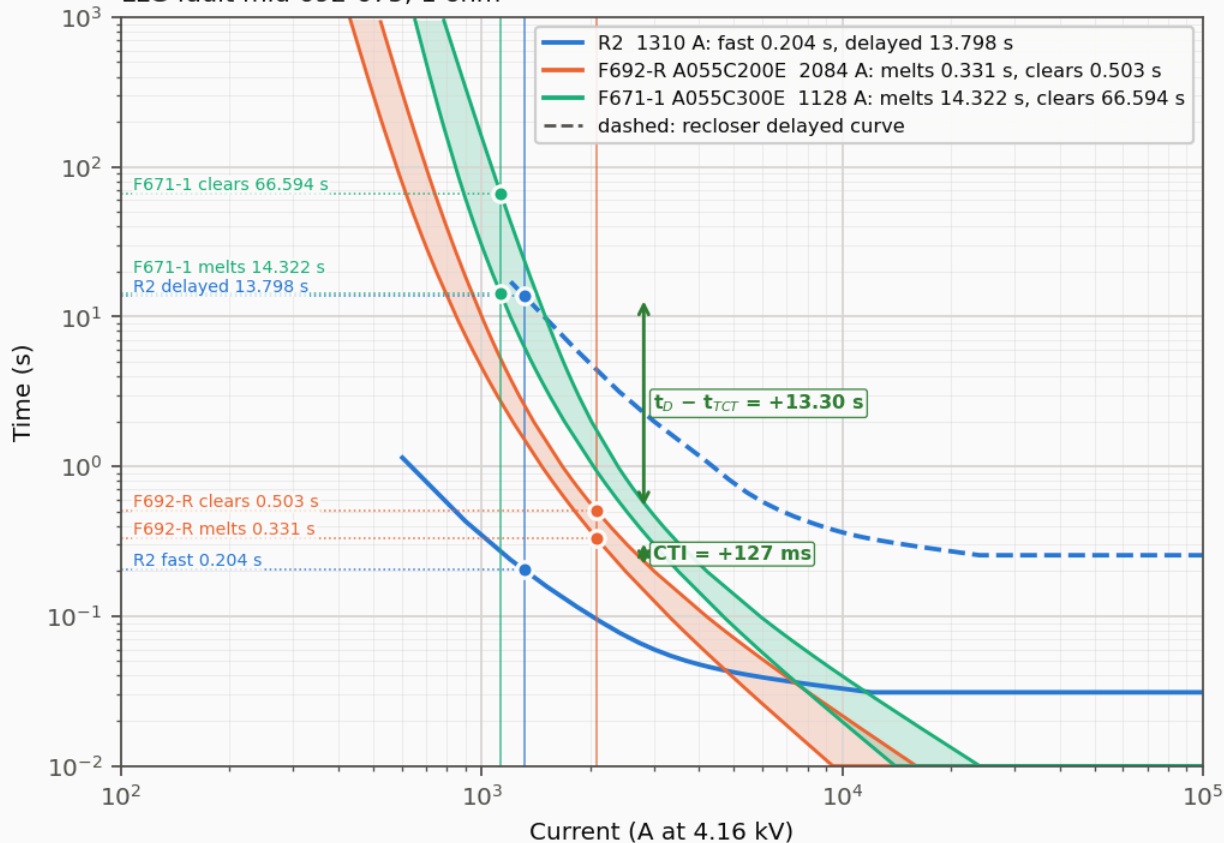
Time-current characteristics of the dual-setting recloser R2: reverse and forward setting



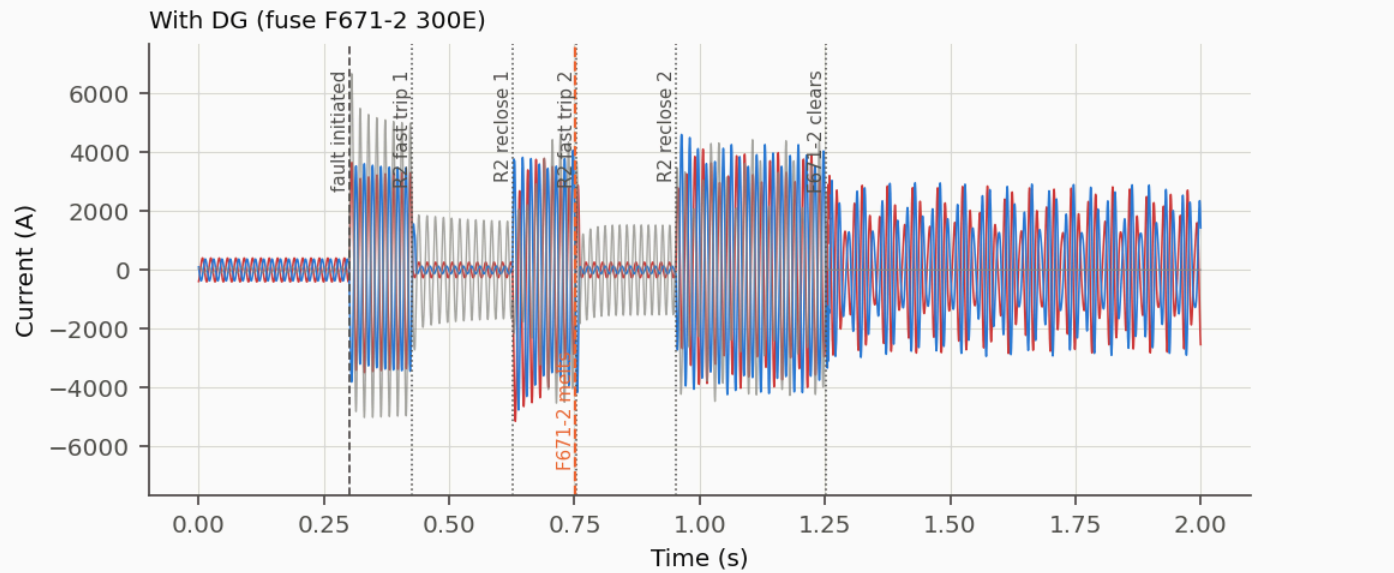
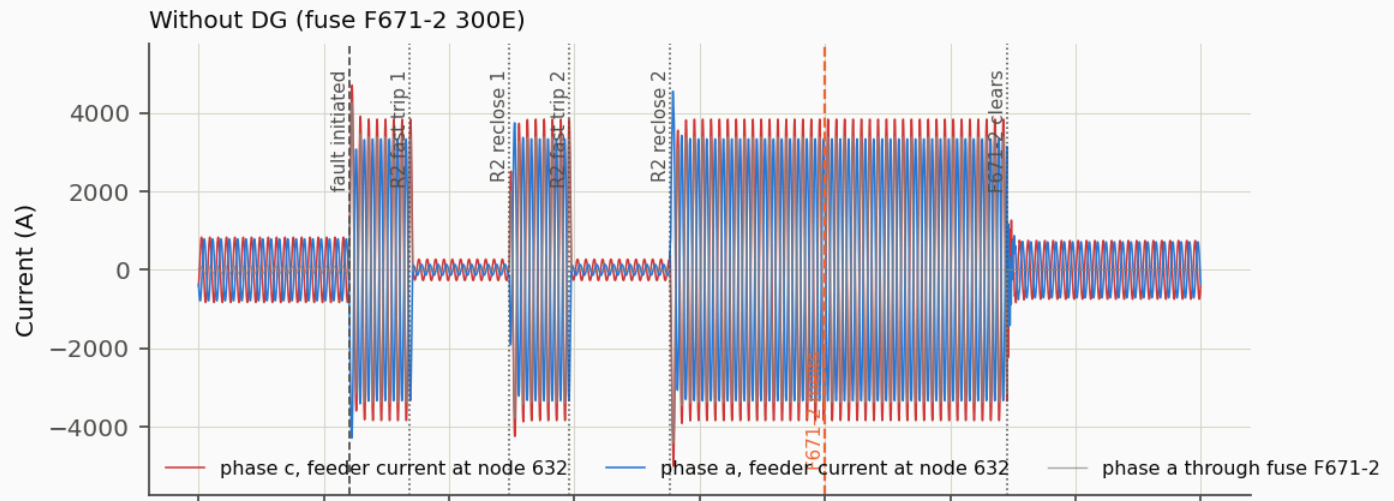
LG fault at 611 - coordination held



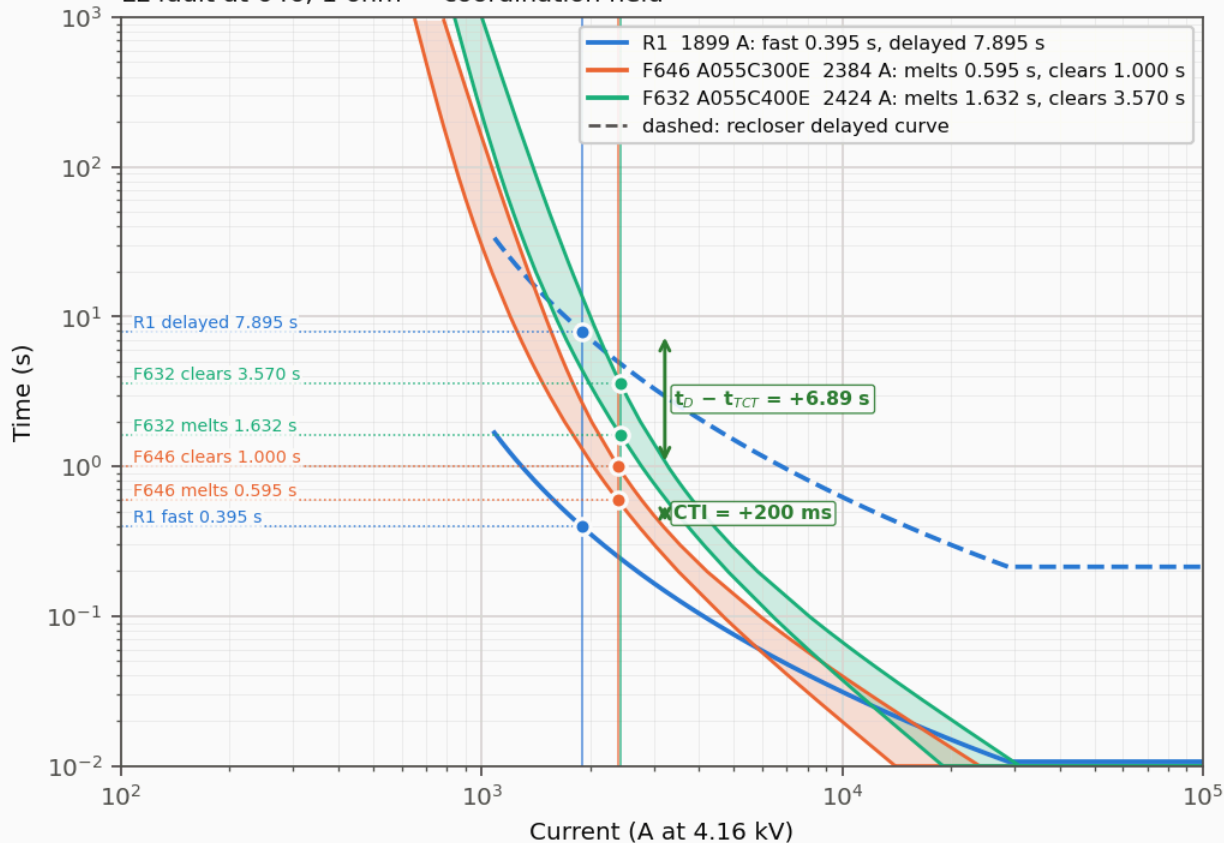
LLG fault mid 692-675, 1 ohm



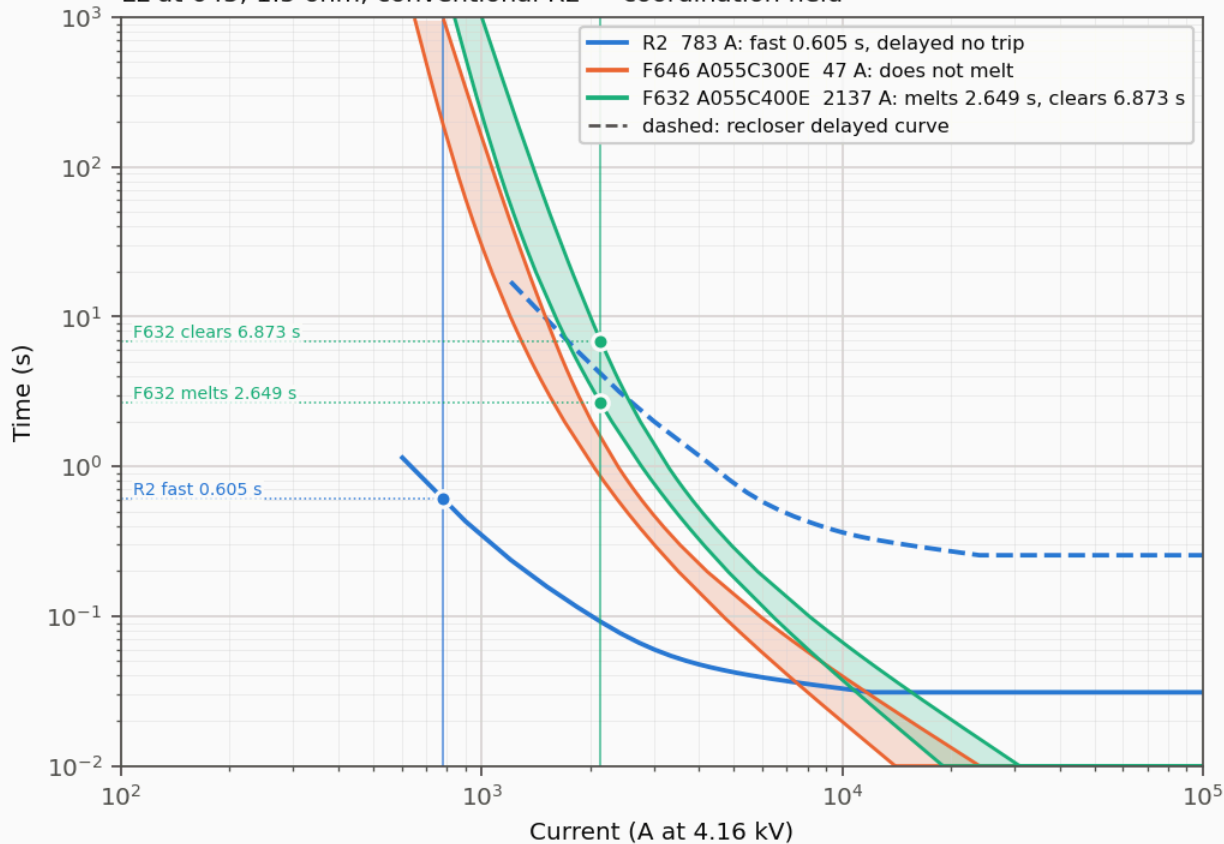
Current at node 632, LL a-c fault at 684 through 0.2 ohm, PowerFactory EMT



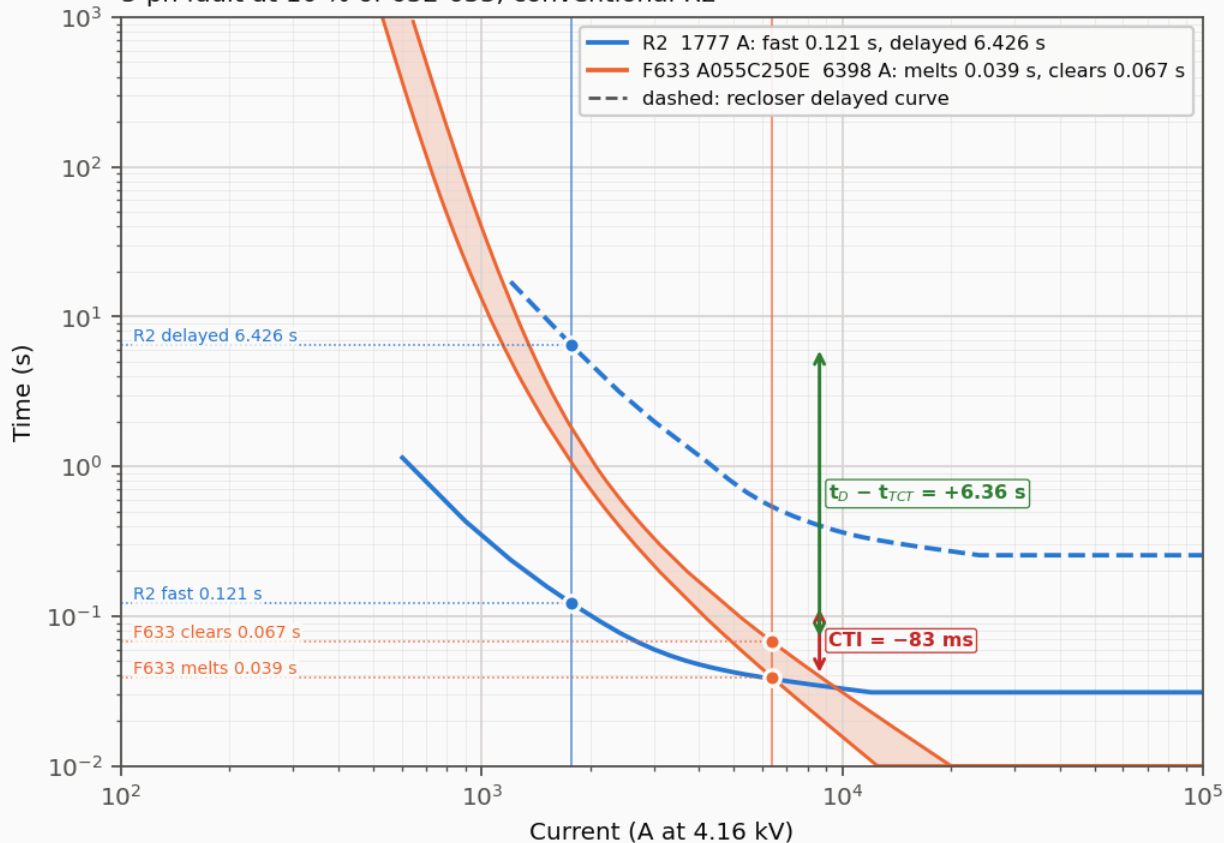
LL fault at 646, 1 ohm - coordination held



LL at 645, 1.5 ohm, conventional R2 - coordination held



3-ph fault at 10 % of 632-633, conventional R2



Coordination without DSDR, DG 4.05 MVA at 692

LG			—										
LL			—								—	—	
LLG			—								—	—	
LLL			—	—	—					—	—	—	
	632	633	634	645	646	671	692	675	680	684	652	611	DL

Faulted node



coordination held

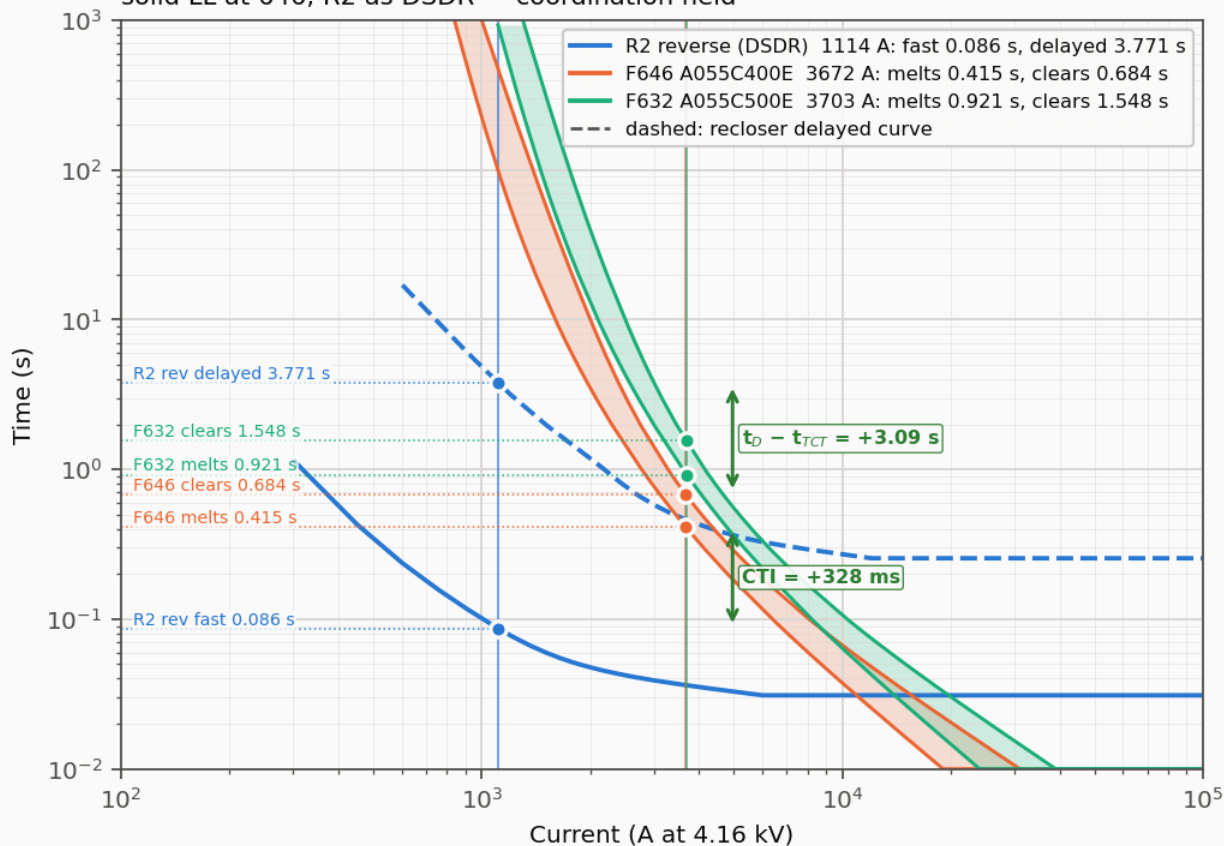


coordination lost

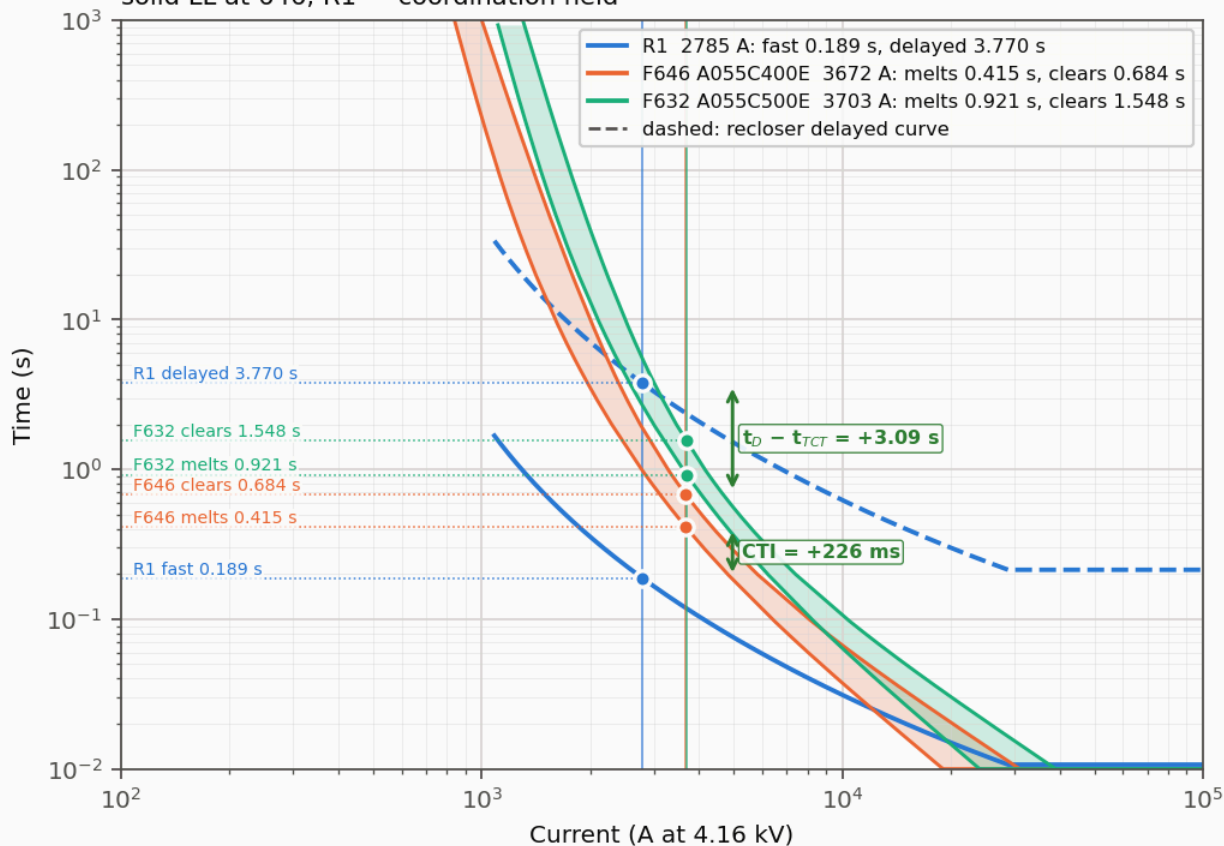


not applicable

solid LL at 646, R2 as DSDR - coordination held



solid LL at 646, R1 - coordination held



Coordination with DSDR, DG 4.05 MVA at 692

LG			—										
LL			—								—	—	
LLG			—								—	—	
LLL			—	—	—					—	—	—	
	632	633	634	645	646	671	692	675	680	684	652	611	DL

Faulted node



coordination held



coordination lost



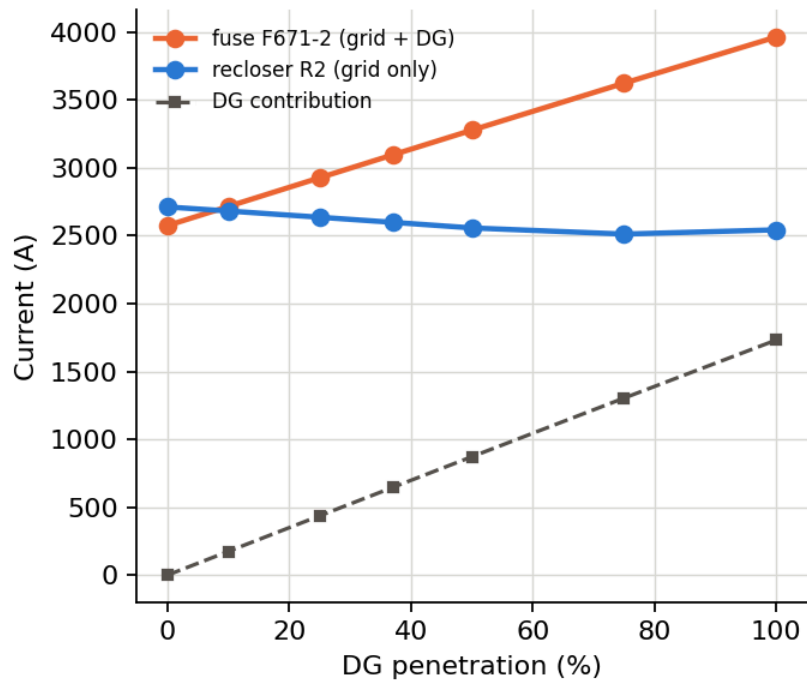
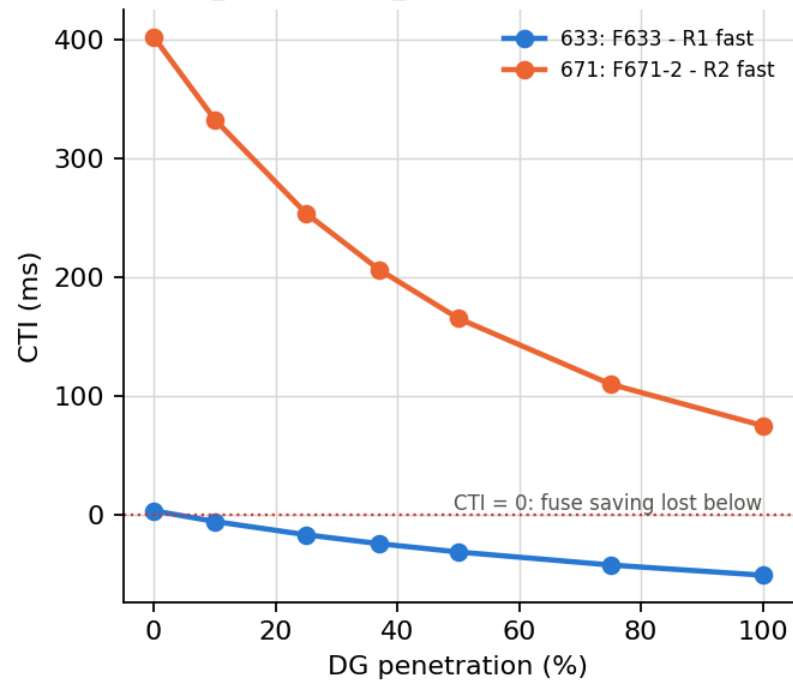
not applicable

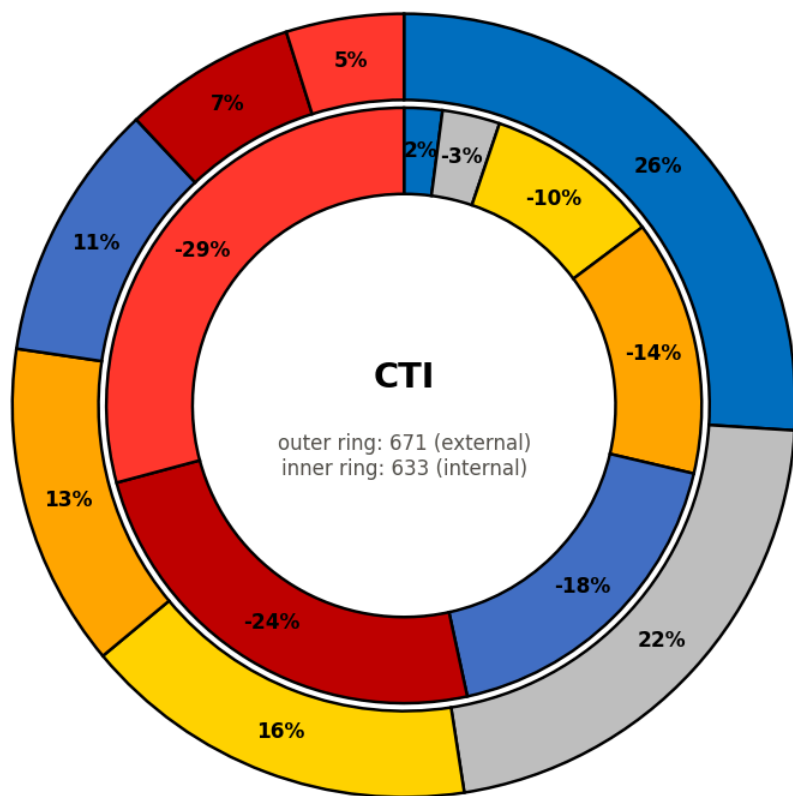


Fault LLL at 633: currents

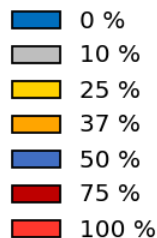


Fault LL at 684 (lateral from 671): currents

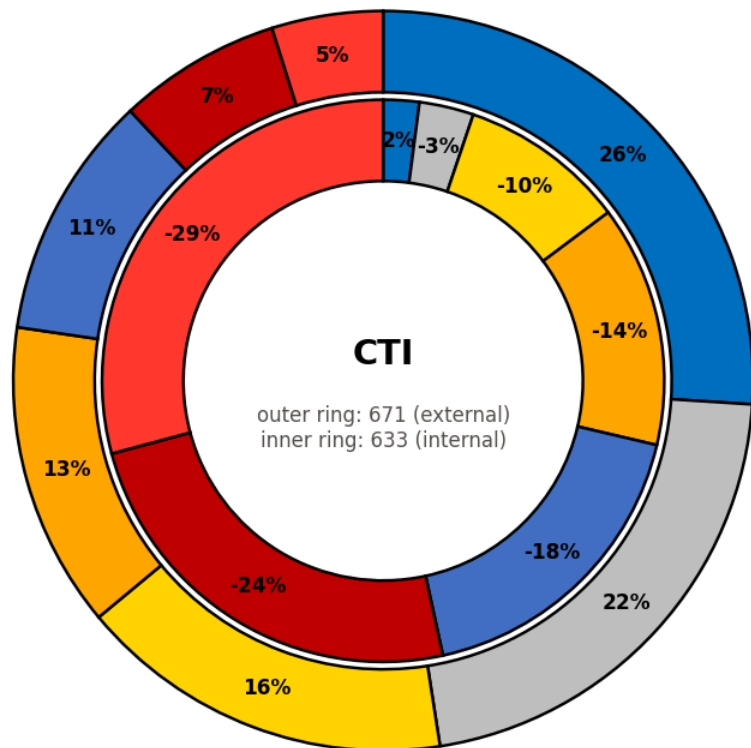
CTI = $t_{\text{MMT}}(\text{fuse}) - t_{\text{fast}}(\text{recloser})$ 



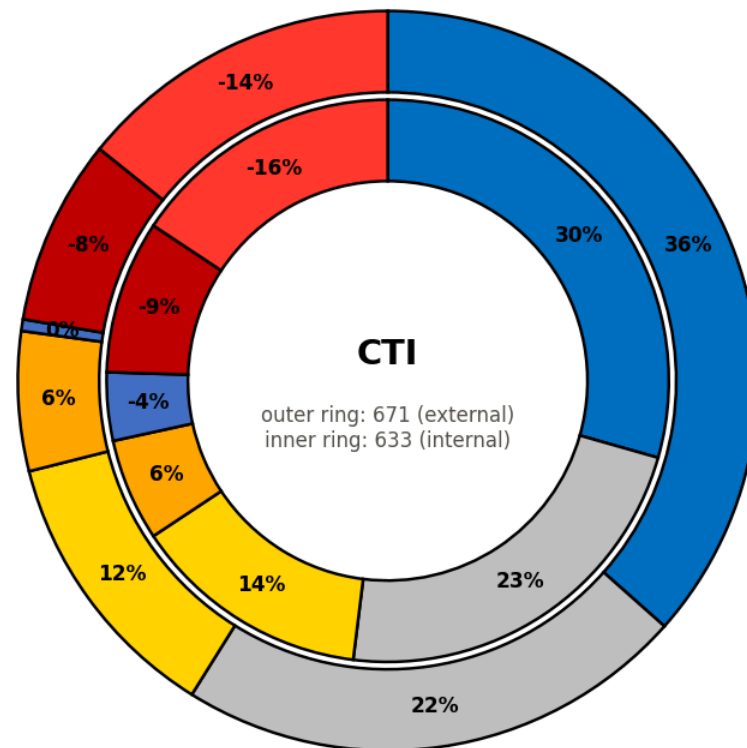
DG penetration



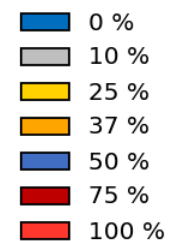
This study (PowerFactory)



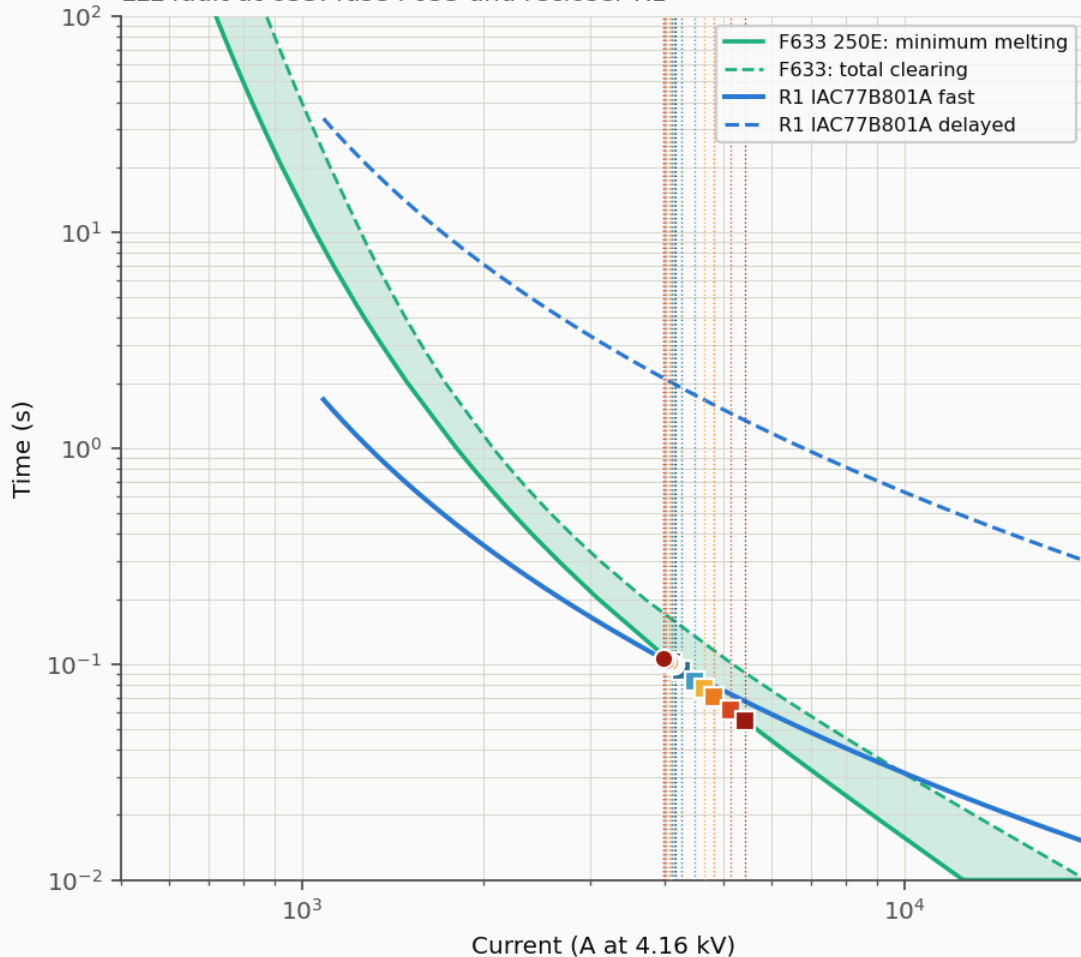
Reference paper (values read from its Fig. 7)



DG penetration



LLL fault at 633: fuse F633 and recloser R1



LL fault at 684 (lateral from 671): fuse F671-2 and recloser R2

